

PAVAN KUMAR U

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Experience

KPIT Technologies Pvt. Ltd

November 2021 – Present

Connected Vehicle Engineer, Verification & Validation

Bangalore, Karnataka

- Delivered end-to-end Verification & Validation across 4 projects spanning CI/CD IVI automation, HIL Instrument Cluster and Infotainment validation, QEMU-based virtualization, and AUTOSAR Classic middleware — for global OEM clients including **Volkswagen Group (CARIAD SE)** and **Stellantis International**.
- Designed, developed, reviewed, and executed 500+ manual and automated test cases across **HIL/SIL** environments using **ECU-TEST TOOL**, **IBM RQM**, **dSPACE ControlDesk** and **Vector CANoe**, reporting 50+ critical defects with full root-cause analysis and requirements traceability.
- Architected CI/CD test automation pipelines using **GitHub Actions and Python (ENNA framework)** on **Goldfish/Cuttlefish** Android Automotive emulators, eliminating manual regression across all build cycles.
- Performed SIL testing using **dSPACE VEOS Player** and automated virtual validation using ECU-TEST, enabling scalable ECU software verification prior to HIL bench integration.
- Led **Level 0 and Level 1 virtualization integration**, deploying client applications on QEMU-based virtual environments with x86, x86-64, and ARM builds, and implemented WebSocket communication layers to validate inter-component integrity.
- Maintained full V-cycle requirements traceability across projects using IBM DOORS, Code Beamer, and IBM RTC — ensuring **ASPICE**-compliant documentation from design through defect closure.

Technical Skills

Programming Languages: Python, C, CAPL

Automotive Protocols: CAN, CAN FD, LIN, UDS (ISO 14229), CANTP, Automotive Ethernet

Validation Tools: ECU-TEST (TraceTronic), Vector CANoe, CANalyzer, dSPACE ControlDesk

Test Environments: SIL, HIL Test Bench, AUTOSAR Classic, Android Automotive (Goldfish / Cuttlefish)

Automotive Domains: Infotainment (IVI), Instrument Cluster, ADAS, BCM

Automation & CI/CD: GitHub Actions, Python (ENNA Framework), PLATIN, CI Pipelines, GCP, JFrog Artifactory, ADB

Requirements & ALM: IBM DOORS / DNG, Code Beamer, IBM RQM, Confluence, Swagger

Defect & Project Tools: JIRA, IBM RTC, Looker Studio, RADAR Dashboards, Firestore, AWS VEW

Standards & Processes: ASPICE, V-Cycle, MISRA C, Agile / SCRUM

Projects

Client: CARIAD SE (Volkswagen Group)

July 2025 – Present

Verification & Validation Engineer — CI/CD IVI Automation

- Architected end-to-end **CI/CD test automation pipeline** for IVI software validation using **GitHub Actions and Python (ENNA framework)** on **Goldfish/Cuttlefish Android** Automotive emulators, eliminating manual regression across Smoke, Sanity, Nightly, and Official build cycles.
- Designed and developed automated test cases for **Infotainment and Navigation** system use-cases using the ENNA framework, supporting functional and system-level validation across Dev and Beta environments.
- Automated ECU communication, system behaviour, and infotainment use-case verification for **HCP3 and ICC ECUs**, ensuring consistent test coverage and early defect detection across release stages.
- Established bench connectivity via PLATIN and integrated test execution results with **RADAR dashboards** via Looker Studio, enabling real-time KPI monitoring and release-gate quality reporting for stakeholders

Client: Stellantis International

July 2023 – June 2025

Verification & Validation Engineer — R2PX Instrument Cluster (HIL)

- Performed end-to-end HIL validation of **Instrument Panel Cluster** ECU covering CAN/CAN FD communication, UDS diagnostics, system alerts, and display logic on a live HIL test bench.
- Analysed Vehicle Function (VF) documents, requirement specifications in IBM DOORS to derive traceable, requirement-mapped test cases for cluster features.
- Designed, developed, reviewed, and executed 500+ manual and automated test cases using IBM RQM and **ECU-TEST TOOL**, achieving comprehensive coverage across communication protocols, display logic, and diagnostic services.
- Performed component-level ECU validation using **Vector CANoe** and CANalyzer to verify CAN/CAN FD protocol behaviour and ECU communication integrity.
- Executed subsystem-level testing using **dSPACE ControlDesk** for HIL system calibration and validation, ensuring accurate signal stimulation and response verification.

- Reported 50+ critical defects via IBM RTC with full root-cause analysis and requirements traceability in IBM DOORS; coordinated resolution with development teams within Agile sprint cycles.

Client: Stellantis International**May 2022 – June 2023***Verification & Validation Engineer — VVPL Virtualization*

- Led virtualization activities for Level 0 and Level 1 integration, deploying and configuring client applications on top of QEMU-based virtual environments to simulate target ECU behaviour.
- Built and configured x86, x86-64, and ARM-based virtual target environments using **QEMU and UNITYAPP** application stack, enabling platform-independent ECU software validation.
- Designed and implemented a WebSocket communication layer to validate inter-process and inter-component communication within the QEMU virtualized environment, ensuring functional integrity of integrated client applications.
- Automated SIL test case execution using **ECU-TEST tool**, enabling repeatable and scalable virtual validation of ECU software across regression and sanity test cycles

Client: KPIT Technologies Pvt. Ltd. — MW Development Project**November 2021 – April 2022***Trainee — AUTOSAR Classic*

- Configured IPCOM and IPCABS AUTOSAR Classic modules from design specifications, ensuring full alignment with project requirements and system architecture definitions.
- Executed unit testing and component testing to verify functional correctness of each module, achieving 100% requirements coverage at module level.
- Enforced MISRA C compliance through systematic static analysis checks, eliminating safety-critical coding violations prior to software integration.
- Established and maintained full V-cycle requirements traceability for IPCOM and IPCABS modules from design through implementation to test closure, achieving zero untraced requirements at sign-off.
- Executed generation tool testing to validate outputs from the AUTOSAR code generation process, ensuring correctness of generated artefacts before integration

Education

Newton's Institute of Engineering, JNTU-Kakinada*Bachelor of Technology in Electronics and Communication | GPA: 7.04***Sri Siddhartha Junior College***Board of Intermediate Education | Marks: 882***Vailankanni Matha Vidyalaya***Class X | GPA: 8.8***Licenses & Certifications**

- **Implementing Software Architecture of a Large-Scale System**

Udemy • Issued July 2023 • Credential ID: UC-cb0fa285-546b-434f-bda7-47164269b119